Mohammad Ammar | Aspiring AI/ML Engineer

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LinkedIn

AI/ML graduate student with hands-on experience in deep learning, reinforcement learning, and computer vision, including CNNs, transformers, and attention-based models. Seeking AI/ML research or applied machine learning internships focused on Agentic AI, LLMs, perception, representation learning, and intelligent systems.

EDUCATION

Khoury College of Computer Sciences, Northeastern University , USA **Sep 2025 – Dec 2027**
Master, Artificial Intelligence
Related courses: Foundation of Artificial Intelligence, Algorithms, Reinforcement Learning

Zakir Husain College of Engineering and Technology, AMU, India **Jul 2020 – Apr 2024**
Bachelor, Computer Engineering
Related courses: Data Structures, Algorithms, Applied Machine Learning

TECHNICAL KNOWLEDGE

AI & Machine Learning:	Python, PyTorch, Hugging Face, Ray RLlib, LangChain, TensorFlow, OpenCV, Pandas, Numpy, Matplotlib, TensorBoard, Pydantic, ChromaDB
Software & Tools:	ROS, Git, Linux, LaTeX, Google Colab, Hydra, Poetry

WORK EXPERIENCE

University of Kiel **Jan 2025 - May 2025**
Machine Learning Research Intern
- Developed MicroCrackAttentionNeXt, a deep learning model for micro-crack detection in high-resolution spatiotemporal images, achieving **95.7% accuracy and 0.957 F1 on imbalanced data**.
- Integrated attention and multi-scale representation learning with manifold-based visualizations to improve model performance and interpretability over CNN and transformer baselines.

University of North Dakota **Jun 2024 – Jan 2025**
Machine Learning Research Intern
- Developed an RL-based UAV framework for GPS spoofing detection and mitigation, achieving **96%+ detection accuracy** and **91%+ task success** in adversarial conditions.
- Implemented a real-time DQN with an optimized experience replay mechanism, delivering **sub-25 ms inference latency** and **6× faster learning**.

Marine Technological Society - Autonomous Underwater Vehicle Club ZHCET **Jul 2022 – May 2023**
Computer Vision Engineer
- Spearheaded the development of custom datasets for vision tasks such as buoy detection and docking station recognition.
- Utilized PyTorch and ONNX to quantize and deploy the YOLO Model on edge device, **gaining 9x speedups** over CPU baseline.

PROJECTS

Fine-Tuning ESM-3 on Soluble High-Enrichment DARPin Sequences **Dec 2025 – Jan 2026**
- Fine-tuned the ESM-3 protein language model on a curated dataset of Designed Ankyrin Repeat Proteins (DARPins).
- Employed LoRA and DoRA PEFT techniques to achieve 98%+ accuracy while keeping the active training parameters <4% of the total model size.

Multiagent Reinforcement Learning Aerial Dogfight **Oct 2025 – Dec 2025**
- Developed a custom 3D dog-fighting environment using PyFlyt, simulating aerial combat scenarios.
- Used Ray RLlib for parallel scalable training and evaluation of the RL agent.
- Used PPO to train two independent agents to learn high level combat maneuvers.

Hybrid Search Engine **June 2025 – Aug 2025**
- Developed a local hybrid search for searching through research articles. Used ChromaDB and Elastic Search to search through the documents based on user's query.
- Used Reciprocal Rank Fusion (RRF) to generate robust results by merging Contextual Search with Keyword Based Search results improving robustness compared to single-method baselines.

PEER-REVIEWED RESEARCH

- Reinforcement Learning Driven Integrated Detection and Mitigation of UAV GPS Spoofing Attacks, IEEE Internet of Things
- Optical Coherence Tomography Image Classification using Light-weight Hybrid Transformers, IEEE
- MicroCrackAttentionNeXt, MDPI Sensors
- From Rules to Pixels: A Decoupled Framework for Segmenting Human-Centric Rule Violations, MusIML Workshop NeurIPS